

FS01 Read Performance

Root-cause investigation into slow file access from the KOR file server — reported April 2026, unresolved until now. The cause is a virtual machine build default, not antivirus, deduplication, or the network.

BOTTOM LINE

The file server is running the wrong disk controller

ROOT CAUSE

KOR-FS01 uses the LSI Logic SAS virtual disk controller. It should be **VMware Paravirtual (PVSCSI)**.

Storage controller	: LSI Adapter, SAS 3000 series, 8-port with 1068	(driver LSI_SAS)
Network adapter	: vmxnet3 Ethernet Adapter, 10 Gbps	← configured correctly
E: cluster size	: 8 KB	← should be 64 KB
Partition offset	: 16,384 KB	← correctly aligned

LSI SAS emulates a mid-2000s hardware controller with a shallow queue depth and high CPU cost per I/O. VMware's own guidance is to use PVSCSI for I/O-intensive workloads — a file server being the textbook case.

The network adapter was built correctly as VMXNET3 at 10 Gbps.

Someone considered the network and left the disk controller at the click-through default. That inconsistency is the signature of a hurried VM build.

This matches the reported timeline exactly: **the problem began when the VMs were rebuilt on a new OS against the same SAN storage.** The disks did not change; the virtual hardware presenting them did.

SYMPTOM

What users experience

Right-clicking a large file on a network share takes up to 45 seconds before the context menu appears. The file icon fails to render beforehand. Small files are unaffected.

The arithmetic

ETABS installer	1,312 MB	
Cold read throughput, client	57 MB/s	
Single full read	23 s	
Explorer reads a large signed EXE twice (icon extraction + Authenticode hash) ×2		
Total	46 s	← matches the reported 45 s

Windows must open an executable to extract its embedded icon, and validating its code signature requires hashing every byte. Two full passes over a 1.3 GB file, at the speed the server can deliver it.

This was raised with the MSP in April 2026 (thread: *"Right Click"*, 7 messages, 14–27 April). Webroot was uninstalled as a test, described as "minimal improvement", and the thread closed without a cause.

EVIDENCE

Measured throughput

PATH	MB/S	INTERPRETATION
Client ← FS01 RAM (warm)	94–98	80% of the client's 1 Gbps NIC — network is healthy
FS01 ← its own E: volume (cold)	74–91	The SAN read path — the constraint
Client ← FS01 disk (cold)	56–58	SAN read plus network, not pipelined

Warm reads are served from the server's RAM and never touch the SAN, which is why they reach 95 MB/s. The moment a file must come off disk, throughput

drops to the storage path's ceiling.

READING THIS CORRECTLY

The client NIC is 1 Gbps (~118 MB/s theoretical). FS01's NIC is 10 Gbps. **The network is not the bottleneck at any point** — a warm read already achieves 80% of the client's link. Every deficit is upstream of the wire.

ELIMINATION

Eight suspects, all ruled out by test

Recorded so none of these is re-investigated.

SUSPECT	TEST PERFORMED	RESULT
Bluebeam context-menu handler	Managed (.NET) handler removed, Explorer restarted	No change
Webroot shell extension	WRShellExt removed	No change
Webroot real-time protection	All shields fully disabled, throughput re-measured	94.0 vs 98.1 MB/s — identical
SMB signing (Server 2025 default)	Get-SmbConnection on live sessions	Signed=False, dialect 3.1.1
Process Monitor filter driver	fltmc filters	Absent (was present in April capture)
Mark-of-the-Web / Internet zone	Registry ZoneMap	UNCAsIntranet=1 already set
Storage faults	60,939 System-log events reviewed	Zero disk, iSCSI or SMB errors
Storport "errors" (6,555)	SCSI sense codes decoded	Sense (0x2, 0x3A, 0x0) = empty optical drive, 1 ms latency

DEDUPLICATION – UNPROVEN, NOT EXONERATED

E: holds **2,798,980 optimised files** saving 10.4 TB. Deduplicated files must be rehydrated from the chunk store on read, converting sequential I/O into random I/O.

This could not be tested. Of 622 large files on the volume, **621 are deduplicated**; the single plain file was pinned in the server's RAM cache. There is no control to measure against. Establishing it would require writing a fresh large file to E: and reading it cold before the optimisation job claims it (default minimum file age: 3 days).

Note the reported problem *predates* deduplication being enabled, so it cannot be the sole cause — but it may be compounding it.

CONFIGURATION

KOR-FS01 as built

Operating system	Windows Server 2025
Disk controller	LSI Adapter, SAS 3000 series (LSI_SAS) – should be PVSCSI
Network adapter	vmxnet3, 10 Gbps – correct
Disks	VMware Virtual disk, BusType SAS, GPT, Fixed provisioning
C: volume	99 GB, 4 KB clusters
E: volume	21 TB (13.1 TB free), NTFS, 8 KB clusters , MFT 4.29 GB
Partition offset	16,384 KB – correctly aligned
Deduplication	Enabled, 2,798,980 files optimised, 10.4 TB saved, last job successful
Dedup schedule	01:45 – 03:50 daily — no working-hours contention

REMEDIATION

What to do

ACTION	EFFORT	RISK	EXPECTED BENEFIT
1. LSI SAS → VMware Paravirtual	One maintenance window; add PVSCSI controller, migrate the data disk, boot	Low — documented by VMware, fully reversible	Highest. Directly addresses the identified cause
2. Re-measure before doing anything else	Minutes — repeat the same read test	None	Confirms the fix and quantifies it
3. 8 KB → 64 KB cluster size on E:	Reformat — requires relocating 8 TB	High — needs a real outage window	Meaningful for large files, but only pursue if step 1 falls short
4. Settle the deduplication question	Write a fresh large file, read cold before optimisation claims it	Low	Determines whether dedup is compounding the problem

SEQUENCE THIS WITH THE WORK ALREADY SCHEDULED

The controller swap needs an FS01 maintenance window — the same window already planned for the EdgeSwitch and pfSense firmware, once off-site replication completes and someone is on site. Three fixes, one outage.

Interim, costing nothing

- Use **Ctrl+C** rather than **right-click** on large files. The keyboard copy builds no context menu, so neither read occurs.
- **robocopy /J** for files of this size — unbuffered I/O, faster than Explorer.
- **Move large installers to the Install share.** It currently holds two files and nobody browses through it, so Explorer never inspects its contents in passing.

METHOD

How this was measured

WinRM and RPC dynamic ports are blocked to KOR machines, so no remote PowerShell was available. All measurement was performed agentlessly:

- **File and log access** over administrative shares (`c$`), including copying `.evtx` files locally for parsing — the only way to read event logs without RPC.
- **Registry inspection** via `reg.exe`, wrapped so RemoteRegistry is started and always restored to Disabled afterwards.
- **Remote execution** by staging a script over `c$` and launching it through a transient service created with `sc.exe` — which works over SMB where RPC does not. Scripts ran as `SYSTEM`; all services were deleted immediately and all temporary files removed.

Throughput figures are sequential reads with a 4 MB buffer, discarding results above 500 MB/s as RAM-cached rather than genuine.

KOR Structural · FS01 read performance investigation · prepared 2026-08-08

Measurements taken from KOR-217 and KOR-FS01 on 2026-08-08. Prior context: MSP thread "*Right Click*", 14–27 April 2026, closed without resolution.